class 5: Data Viz with ggplot

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Background

There are lot's of ways to make plots in R. These include so-called "base R" (like the `plot()`) and add on packages like **ggplot**,

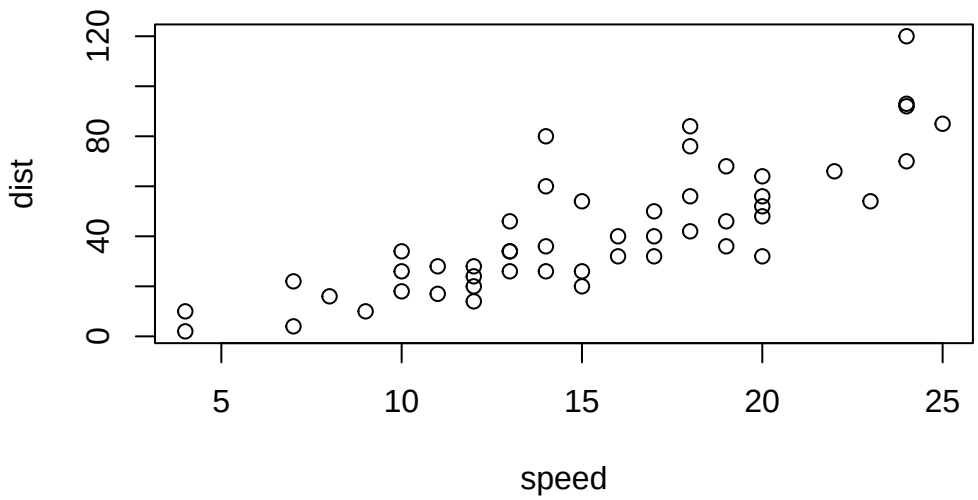
Let's make the same plot with these two graphics systems. We can use the inbuilt `car` dataset:

```
head( cars )
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

With "base R" we can simply:

```
plot (cars)
```



Now let's try ggplot. First i need to install the package using `install.packages("ggplot2")`

> **N.B.** We never run on `install.packages("ggplot2")`in a code chunk otherwise we will re-install needlesly every time we render our document.

Every time we want to use an add-on package we need to load it up with a call to `library()`

```
library(ggplot2)
```

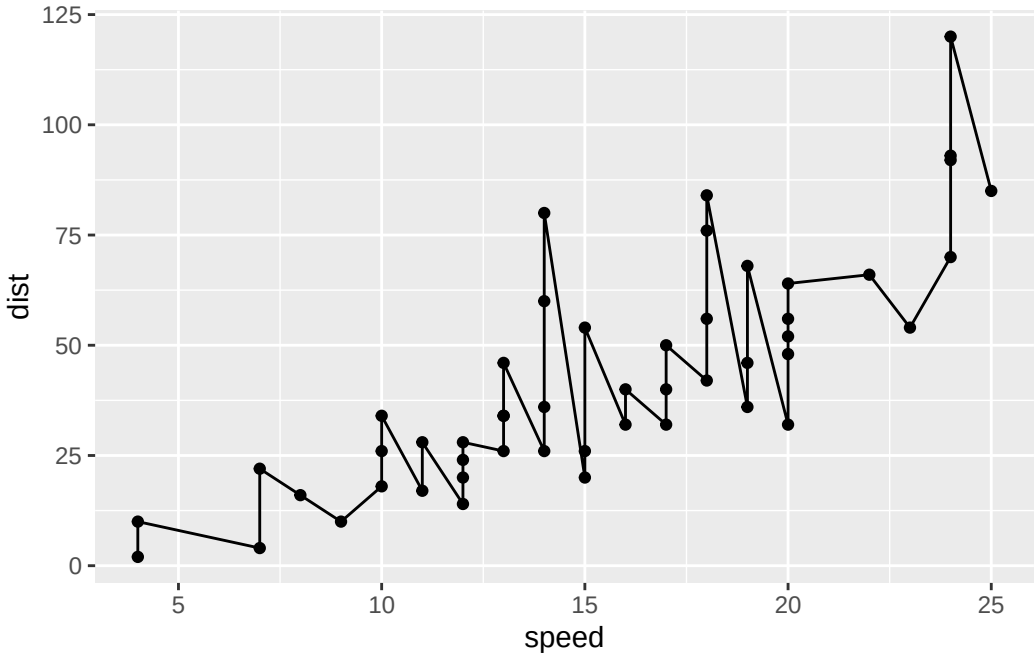
```
Warning: package 'ggplot2' was built under R version 4.5.2
```

```
ggplot(cars)
```

Every ggplot needs at least 3 things:

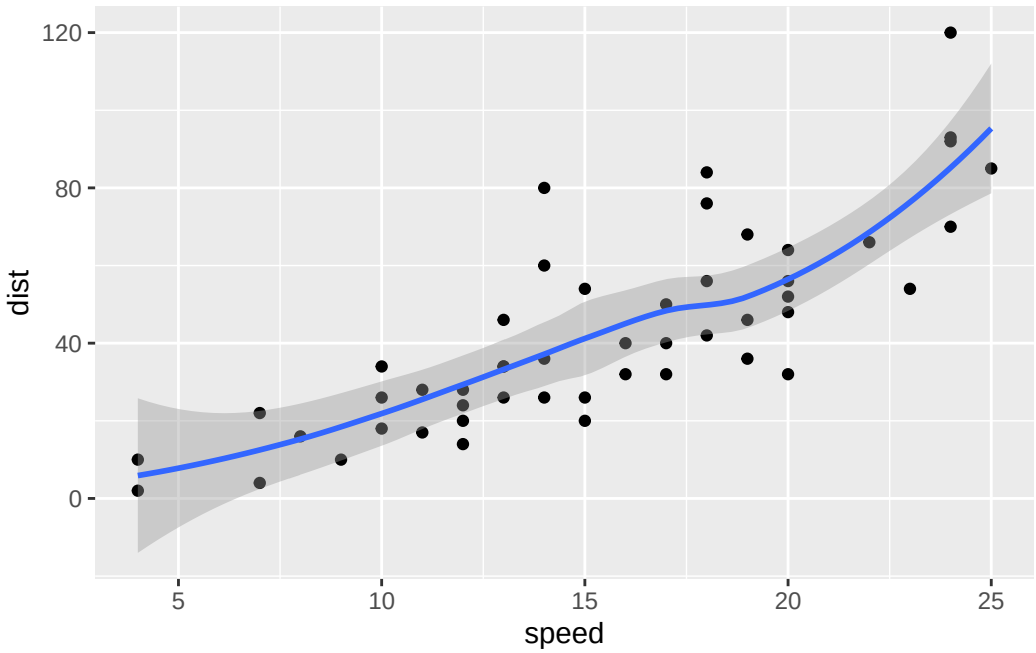
1. The **data** i.e. stuff to plot as a data.frame
2. The **aes** or aesthetics that map the data to the plot
3. The **geom_** or geometry i.e. the plot type such as points,lines etc.

```
ggplot(cars) +
  aes(x=speed, y=dist) +
geom_point() +
  geom_line()
```



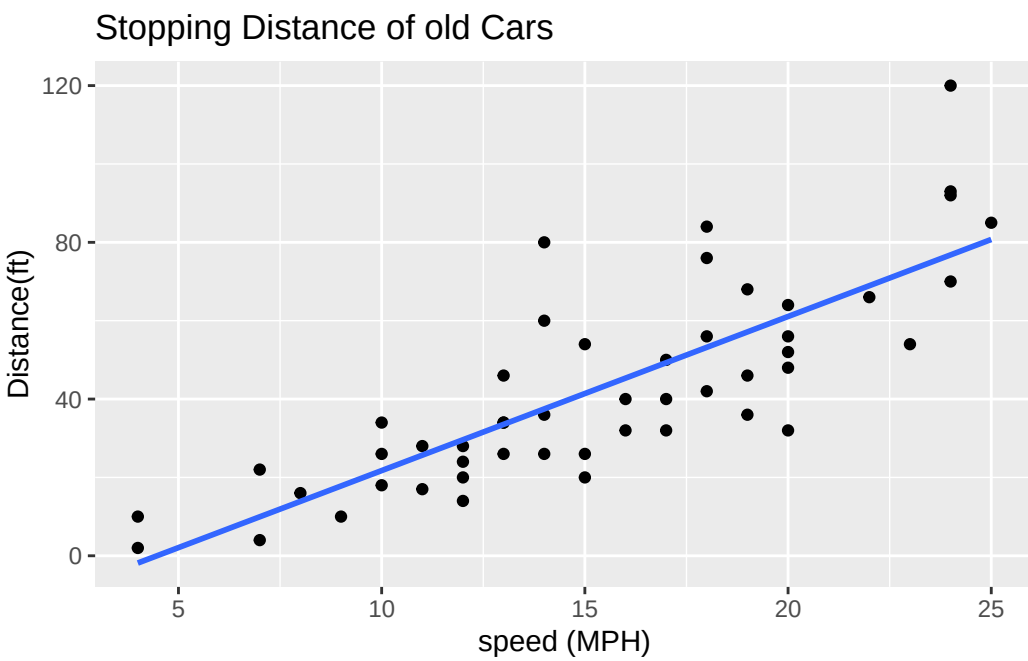

```
ggplot(cars) +
  aes(x=speed, y=dist) +
geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method= "lm", se= FALSE) + labs(x="speed (MPH)", y= "Distance(ft)", title= "St
```

```
`geom_smooth()` using formula = 'y ~ x'
```



###Gene Expression Plot

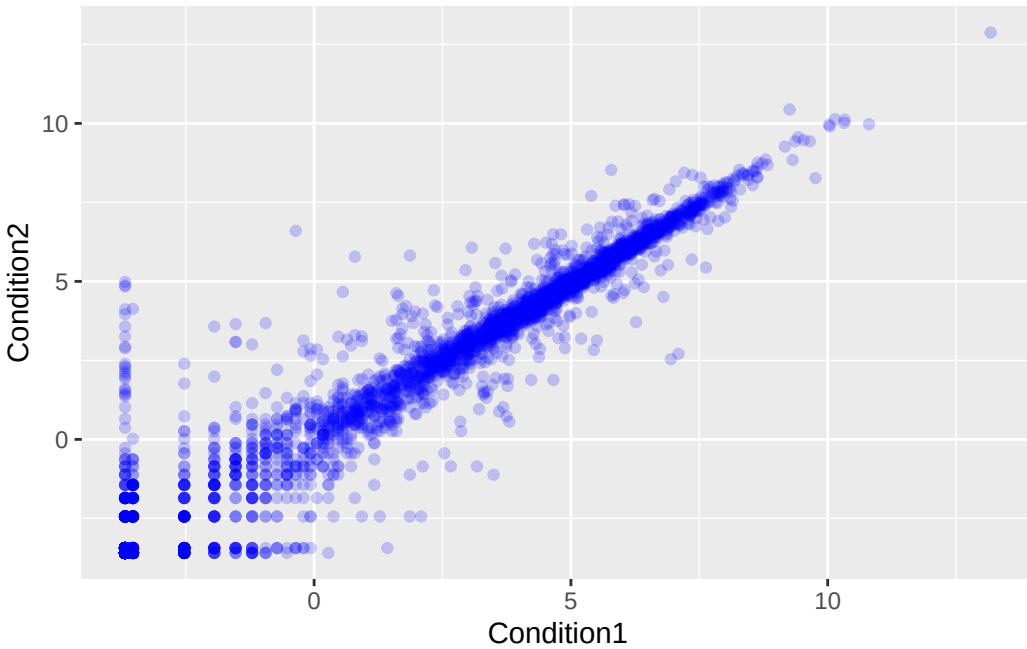
Read some data on the effects of BLP-1 inhibitor (drug) on gene expression values:

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Version 1 plot - start simple by getting some ink on the page.

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2,) +
  geom_point(col="blue", alpha=0.2)
```

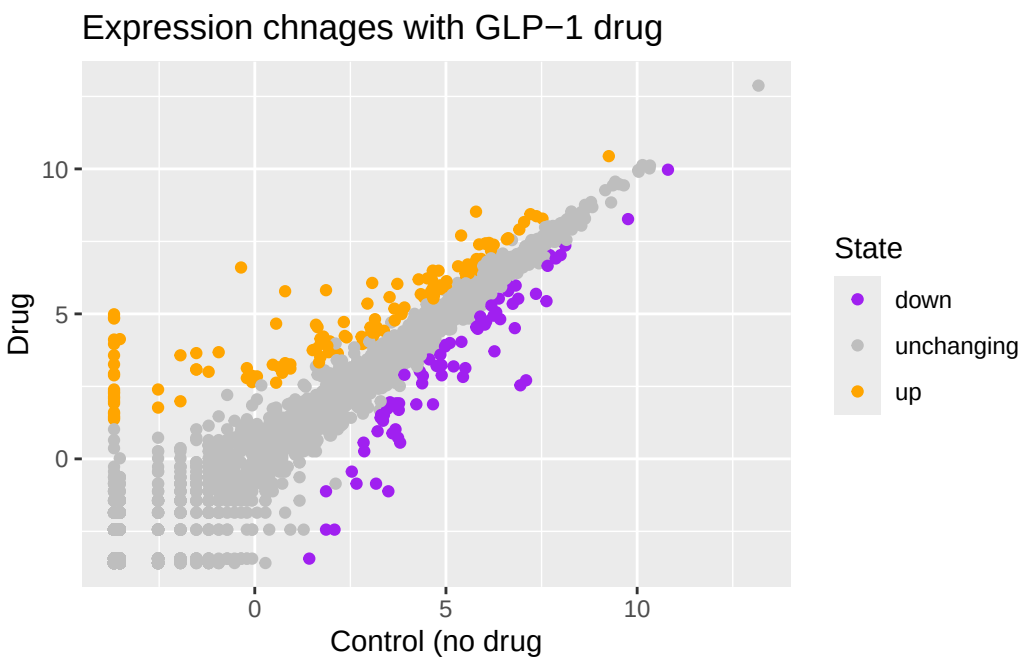


Let's color by `State` up, down or not change.

```
table( genes$State )
```

```
      down unchanging         up
        72       4997        127
```

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  scale_color_manual(values= c("purple",
                               "grey",
                               "orange"))+
labs(x= "Control (no drug",
     y= "Drug",
     title= "Expression chnages with GLP-1 drug")
```




```
theme_classic()
```

```
<theme> List of 144
 $ line                            : <ggplot2::element_line>
  ..@ colour       : chr "black"
  ..@ linewidth    : num 0.5
  ..@ linetype     : num 1
  ..@ lineend      : chr "butt"
  ..@ linejoin     : chr "round"
  ..@ arrow        : logi FALSE
  ..@ arrow.fill   : chr "black"
  ..@ inherit.blank: logi TRUE
 $ rect                            : <ggplot2::element_rect>
  ..@ fill         : chr "white"
  ..@ colour       : chr "black"
  ..@ linewidth    : num 0.5
  ..@ linetype     : num 1
  ..@ linejoin     : chr "round"
  ..@ inherit.blank: logi TRUE
 $ text                            : <ggplot2::element_text>
  ..@ family       : chr ""
  ..@ face         : chr "plain"
```

```
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : chr "black"
 ..@ size         : num 11
 ..@ hjust        : num 0.5
 ..@ vjust        : num 0.5
 ..@ angle        : num 0
 ..@ lineheight   : num 0.9
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 0
 ..@ debug        : logi FALSE
 ..@ inherit.blank: logi TRUE
$ title                          : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : NULL
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ point                          : <ggplot2::element_point>
 ..@ colour       : chr "black"
 ..@ shape        : num 19
 ..@ size         : num 1.5
 ..@ fill         : chr "white"
 ..@ stroke       : num 0.5
 ..@ inherit.blank: logi TRUE
$ polygon                        : <ggplot2::element_polygon>
 ..@ fill         : chr "white"
 ..@ colour       : chr "black"
 ..@ linewidth    : num 0.5
 ..@ linetype     : num 1
 ..@ linejoin     : chr "round"
 ..@ inherit.blank: logi TRUE
$ geom                           : <ggplot2::element_geom>
 ..@ ink        : chr "black"
```

```
 ..@ paper      : chr "white"
 ..@ accent     : chr "#3366FF"
 ..@ linewidth  : num 0.5
 ..@ borderwidth: num 0.5
 ..@ linetype   : int 1
 ..@ bordertype : int 1
 ..@ family     : chr ""
 ..@ fontsize   : num 3.87
 ..@ pointsize  : num 1.5
 ..@ pointshape : num 19
 ..@ colour     : NULL
 ..@ fill       : NULL
$ spacing                         : 'simpleUnit' num 5.5points
 ..- attr(*, "unit")= int 8
$ margins                         : <ggplot2::margin> num [1:4] 5.5 5.5 5.5 5.5
$ aspect.ratio                    : NULL
$ axis.title                      : NULL
$ axis.title.x                    : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 2.75 0 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.title.x.top                : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 0
 ..@ angle        : NULL
```

```
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.75 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.title.x.bottom             : NULL
$ axis.title.y                    : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : num 90
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.75 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.title.y.left               : NULL
$ axis.title.y.right              : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : num -90
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.75
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text                       : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
```

```
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 ..@ hjust        : NULL
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : NULL
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.x                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 1
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 2.2 0 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.x.top                 : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
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 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : NULL
 ..@ vjust        : num 0
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 2.2 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.x.bottom              : NULL
$ axis.text.y                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
```

```
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 1
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 0
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.y.left                : NULL
$ axis.text.y.right               : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 0
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 0 0 2.2
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ axis.text.theta                 : NULL
$ axis.text.r                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 0.5
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : <ggplot2::margin> num [1:4] 0 2.2 0 2.2
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
```

```
$ axis.ticks                      : <ggplot2::element_line>
 ..@ colour       : NULL
 ..@ linewidth    : NULL
 ..@ linetype     : NULL
 ..@ lineend      : NULL
 ..@ linejoin     : NULL
 ..@ arrow        : logi FALSE
 ..@ arrow.fill   : NULL
 ..@ inherit.blank: logi TRUE
$ axis.ticks.x                    : NULL
$ axis.ticks.x.top                : NULL
$ axis.ticks.x.bottom             : NULL
$ axis.ticks.y                    : NULL
$ axis.ticks.y.left               : NULL
$ axis.ticks.y.right              : NULL
$ axis.ticks.theta                : NULL
$ axis.ticks.r                    : NULL
$ axis.minor.ticks.x.top          : NULL
$ axis.minor.ticks.x.bottom       : NULL
$ axis.minor.ticks.y.left         : NULL
$ axis.minor.ticks.y.right        : NULL
$ axis.minor.ticks.theta          : NULL
$ axis.minor.ticks.r              : NULL
$ axis.ticks.length               : 'rel' num 0.5
$ axis.ticks.length.x             : NULL
$ axis.ticks.length.x.top         : NULL
$ axis.ticks.length.x.bottom      : NULL
$ axis.ticks.length.y             : NULL
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$ axis.ticks.length.y.right       : NULL
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$ axis.ticks.length.r             : NULL
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$ axis.minor.ticks.length.x.bottom: NULL
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$ axis.minor.ticks.length.y.left  : NULL
$ axis.minor.ticks.length.y.right : NULL
$ axis.minor.ticks.length.theta   : NULL
$ axis.minor.ticks.length.r       : NULL
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```

```
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 ..@ inherit.blank: logi TRUE
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$ axis.line.x.top                 : NULL
$ axis.line.x.bottom              : NULL
$ axis.line.y                     : NULL
$ axis.line.y.left                : NULL
$ axis.line.y.right               : NULL
$ axis.line.theta                 : NULL
$ axis.line.r                     : NULL
$ legend.background               : <ggplot2::element_rect>
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 ..@ colour       : logi NA
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 ..@ linetype     : NULL
 ..@ linejoin     : NULL
 ..@ inherit.blank: logi TRUE
$ legend.margin                   : NULL
$ legend.spacing                  : 'rel' num 2
$ legend.spacing.x                : NULL
$ legend.spacing.y                : NULL
$ legend.key                      : NULL
$ legend.key.size                 : 'simpleUnit' num 1.2lines
 ..- attr(*, "unit")= int 3
$ legend.key.height               : NULL
$ legend.key.width                : NULL
$ legend.key.spacing              : NULL
$ legend.key.spacing.x            : NULL
$ legend.key.spacing.y            : NULL
$ legend.key.justification        : NULL
$ legend.frame                    : NULL
$ legend.ticks                    : NULL
$ legend.ticks.length             : 'rel' num 0.2
$ legend.axis.line                : NULL
$ legend.text                     : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
```

```
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : 'rel' num 0.8
 ..@ hjust        : NULL
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : NULL
 ..@ debug        : NULL
 ..@ inherit.blank: logi TRUE
$ legend.text.position            : NULL
$ legend.title                    : <ggplot2::element_text>
 ..@ family       : NULL
 ..@ face         : NULL
 ..@ italic       : chr NA
 ..@ fontweight   : num NA
 ..@ fontwidth    : num NA
 ..@ colour       : NULL
 ..@ size         : NULL
 ..@ hjust        : num 0
 ..@ vjust        : NULL
 ..@ angle        : NULL
 ..@ lineheight   : NULL
 ..@ margin       : NULL
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 ..@ inherit.blank: logi TRUE
$ legend.title.position           : NULL
$ legend.position                 : chr "right"
$ legend.position.inside          : NULL
$ legend.direction                : NULL
$ legend.byrow                    : NULL
$ legend.justification            : chr "center"
$ legend.justification.top        : NULL
$ legend.justification.bottom     : NULL
$ legend.justification.left       : NULL
$ legend.justification.right      : NULL
$ legend.justification.inside     : NULL
 [list output truncated]
@ complete: logi TRUE
@ validate: logi TRUE
```

Going further with gapmider

Here we explore the famous `gapminder` data sets with some custom plots.

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.ts

gapminder <- read.delim(url)
head(gapminder)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
4 Afghanistan      Asia 1967  34.020 11537966  836.1971
5 Afghanistan      Asia 1972  36.088 13079460  739.9811
6 Afghanistan      Asia 1977  38.438 14880372  786.1134
```

Q. How many rows does this data set have?

```
nrow(gapminder)
```

```
[1] 1704
```

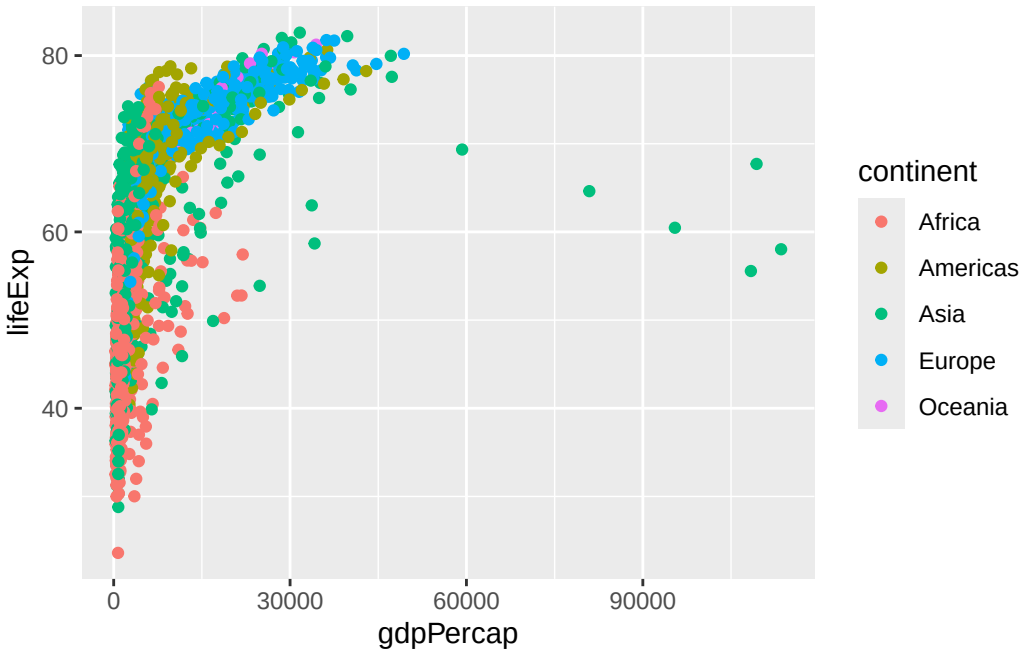
How many different continents are in this date set?

```
table(gapminder$continent)
```

```
  Africa Americas     Asia   Europe  Oceania 
     624      300      396      360       24 
```

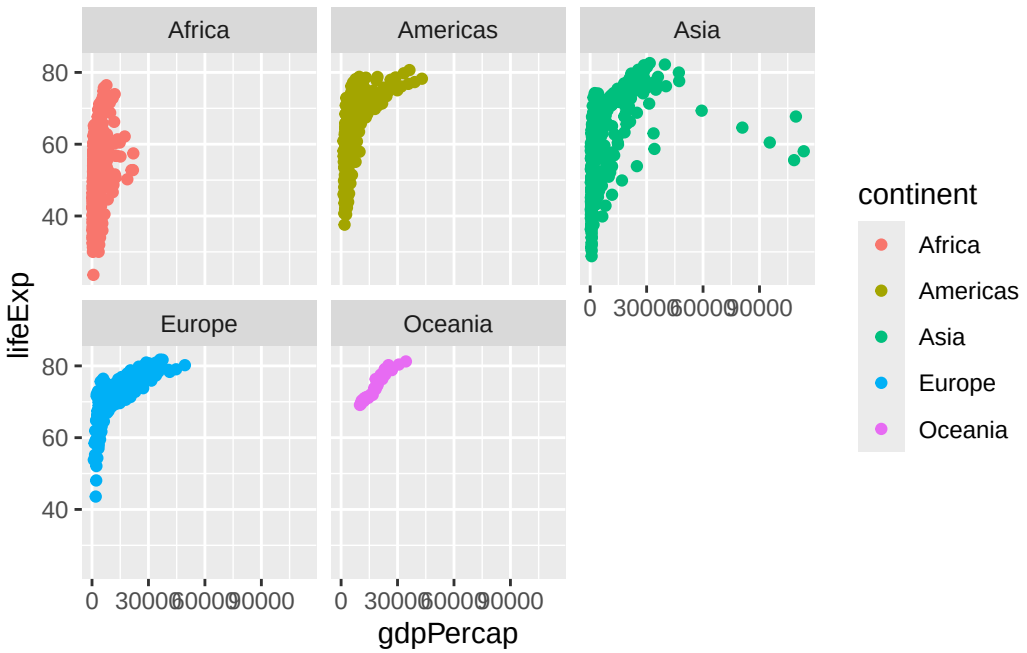
version 1 plot gdppercap vs LifeExp for all rows

```
ggplot (gapminder) +
  aes(gdpPercap, lifeExp, col= continent) +
  geom_point()
```



I want to see a plot for each continents - in ggplot lingo this is called "faceting"

```
ggplot (gapminder) +
  aes(gdpPercap, lifeExp, col= continent) +
  geom_point() +
  facet_wrap(~ continent)
```



First look at the dplyr package

Another add-on package with a function called `filter()` that we want to use.

```
library(dplyr)
```

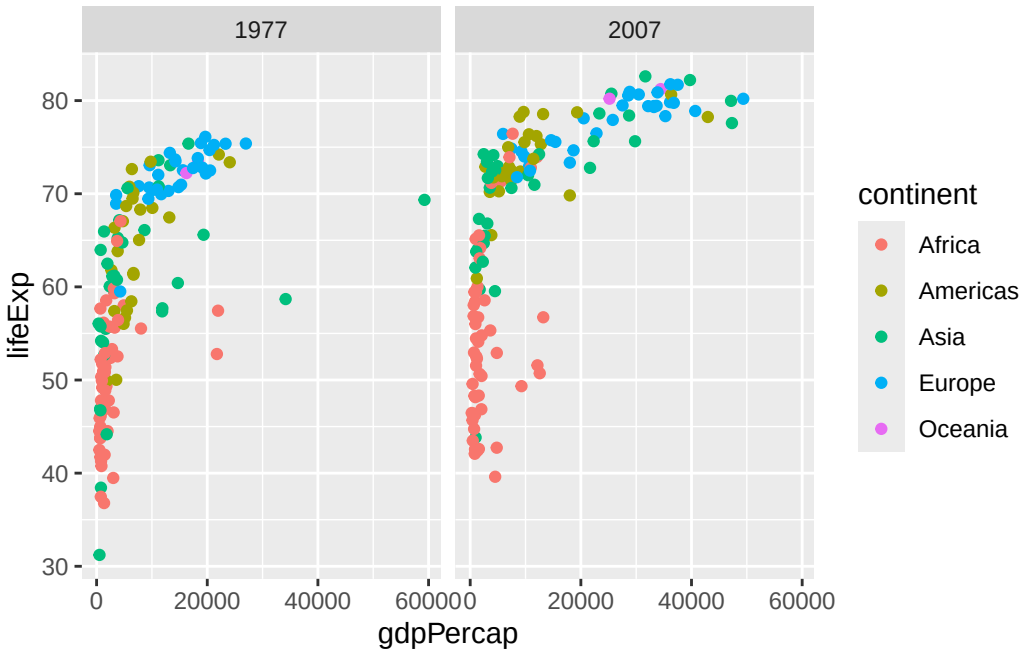
```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```
input <-filter(gapminder, year==2007 | year==1977)
library(ggplot2)
ggplot (input) +
```

```
aes(gdpPercap, lifeExp, col= continent) +
geom_point() +
facet_wrap(~ year)
```